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The importance of normalization

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Normalizing databases is an essential requirement to have a clean and optimized database structure. It helps prevent repeating data, lets you have consistent and accurate data, and improves the overall structure and organization of the complete database. The overall benefit of the normalization process makes every database easy to navigate and improves the overall efficiency, because it ensures no duplicate data, and proper relations between the entities.

Another major advantage of normalization is data integrity. Since each piece of information is stored in its appropriate location, the database can enforce stronger relationships between entities through primary keys and foreign keys. This reduces the possibility of invalid references, orphan records, or contradictory data within the system.

Normalization also contributes to better performance in long-term projects. Even though some highly normalized databases may require more joins, the reduction of redundant information often decreases storage usage and simplifies future modifications to the system. This becomes especially important in scalable applications where databases continuously grow in size and complexity.

Now with generative AI models this process can be simplified. It can help database developers normalize things faster or correct the normalization they already applied. Let's say that the developer already normalized the database, then he passes it to the AI. The AI can tell him that there is a table that can be erased as it has redundant data correlating to another table.

This is very useful, as maybe this extra table could create duplicate data that is not needed or shows data that isn't needed in that specific table or a view related to it. This makes it faster to catch errors within the database before it starts giving real problems when accessing it. By detecting these issues early, developers can prevent larger problems related to performance, maintenance, and data consistency before the database is deployed in a real environment.

However, AI will not replace real developers. Database developers fully understand how the system works, how the tables are connected, and the complete context behind the architecture and business logic. Although AI can analyze database structures and provide recommendations, it may still produce mistakes or misunderstand certain design decisions. Because of this, AI should be viewed as a support tool that assists developers rather than a replacement for them.